wound and unwound string probes will be inversely related to one another.4

If string theory describes our universe, why have we not encountered these two possible notions of distance in any of our day-to-day or scientific endeavors? Any time we talk about distance, we do so in a manner that conforms to our experience of there being one concept of distance without any hint of there being a second notion. Why have we missed the alternative possibility? The answer is that although there is a high degree of symmetry in our discussion, whenever R (and hence 1/R as well) differ significantly from the value 1 (meaning, again, 1 times the Planck length), then one of our operational definitions proves extremely difficult to carry out while the other proves extremely easy to carry out. In essence, we have always carried out the easy approach, completely unaware of there being another possibility.

The discrepancy in difficulty between the two approaches is due to the very different masses of the probes used—high-winding-energy/low-vibration-energy, and vice versa—if the radius R (and hence 1/R as well) differs significantly from the Planck length (that is, R = 1). “High” energy here, for radii that are vastly different from the Planck length, corresponds to incredibly massive probes—billions and billions of times heavier than the proton, for instance—while “low” energy corresponds to probe masses at most a speck above zero. In such circumstances, there is a monumental difference in difficulty between the two approaches, since even producing the heavy-string configurations is an undertaking that, at present, is beyond our technological prowess. In practice, then, only one of the two approaches is technologically feasible—the one involving the lighter of the two types of string configurations. This is the one used implicitly in all of our discussions involving distance encountered to this point. This is the one that informs and hence meshes with our intuition.

Putting issues of practicality aside, in a universe governed by string theory one is free to measure distances using either of the two approaches. When astronomers measure the “size of the universe” they do so by examining photons that have traveled across the cosmos and have happened to enter their telescopes. No pun intended, photons are